

Conditional Expectation

One picture carries the whole lecture: the table of outcomes of two dice, Y the first, Z the second, $X = Y + Z$. The rows of the table are the events $\{Y = y\}$.

1. **Given a value:** $\mathbb{E}[X \mid Y = y]$ The average of X along row y . A number for each y , hence a function of y .
2. **Law of total expectation** The average of the row averages, weighted by row probabilities, is the overall average.
3. **Given the variable:** $\mathbb{E}[X \mid Y]$ Paint every row with its own average: a new random variable.
4. **Properties** Tower, taking out what is known, independence. Each is a one-glance fact about rows.
5. **Two applications** Random sums, and waiting times: hard head-on, one line by conditioning.

Three Problems We Cannot Yet Do

Each asks for an expected value. With the tools so far, the only route is to find the distribution of the quantity first.

Problem A: a random number of coins

Roll a die, getting N . Toss N fair coins; H is the number of heads. Find $\mathbb{E}[H]$.

Head-on: $p_H(h) = \sum_{n=\max(h,1)}^6 \frac{1}{6} \binom{n}{h} 2^{-n}$ for $h = 0, \dots, 6$, seven sums, then $\sum_h h p_H(h)$.

Problem B: an insurer's payout

An insurer receives N claims in a year, $\mathbb{E}[N] = 20$, each of mean size £1500, sizes independent of the count. Find the expected total payout $S = X_1 + \dots + X_N$.

Head-on: the distribution of S mixes, over n , the n -fold convolution of the claim-size distribution, and neither distribution was given.

Problem C: waiting for HH

Toss a fair coin until two heads in a row appear; T is the number of tosses used. Find $\mathbb{E}[T]$.

Head-on: $\mathbb{P}(T = n) = F_{n-1}/2^n$ with $F_1, F_2, F_3, \dots = 1, 1, 2, 3, 5, \dots$ the Fibonacci numbers, then $\sum_n n \mathbb{P}(T = n)$.

All three are solved by conditioning on the first stage: slides 12, 31, 34.

The Table

Two fair dice: $(\Omega, \mathcal{F}, \mathbb{P})$ with $\Omega = \{1, \dots, 6\}^2$, $\mathcal{F}$ all subsets of Ω , and $\mathbb{P}(\{\omega\}) = \frac{1}{36}$ for each outcome $\omega = (y, z)$.

- $Y: \Omega \rightarrow \{1, \dots, 6\}$, $Y(\omega) = y$, the first die: the **row**.
- $Z: \Omega \rightarrow \{1, \dots, 6\}$, $Z(\omega) = z$, the second die: the **column**.
- $X: \Omega \rightarrow \{2, \dots, 12\}$, $X(\omega) = y + z$, the sum: the **number in the cell**.

Shorthand, used throughout:

$\{Y = y\} := \{\omega \in \Omega : Y(\omega) = y\}$ and
 $\mathbb{P}(Y = y) := \mathbb{P}(\{Y = y\})$; likewise $\{X \in A\}$,
 $\mathbb{P}(X \leq x)$, $\mathbb{P}(X = x, Y = y)$.

Every random variable on this Ω writes a number in each of the 36 cells; every event is a set of cells, with probability the number of cells over 36.

$$X(\omega) = y + z$$

$z =$	1	2	3	4	5	6
$y = 1$	2	3	4	5	6	7
$y = 2$	3	4	5	6	7	8
$y = 3$	4	5	6	7	8	9
$y = 4$	5	6	7	8	9	10
$y = 5$	6	7	8	9	10	11
$y = 6$	7	8	9	10	11	12

Conditioning on $Y = y$ Is Restricting to a Row

The event $\{Y = 4\}$ is the fourth row: six cells, probability $\frac{6}{36} = \frac{1}{6}$. Given $Y = 4$, the outcome lies in that row, and each of its six cells has conditional probability

$$\mathbb{P}(\{\omega\} \mid Y = 4) = \frac{\mathbb{P}(\{\omega\} \cap \{Y = 4\})}{\mathbb{P}(Y = 4)} = \frac{1/36}{1/6} = \frac{1}{6}.$$

So, given $Y = 4$, X takes each of the values 5, 6, ..., 10 with probability $\frac{1}{6}$: the distribution of the numbers in row 4.

row 4 is the event $\{Y = 4\}$

$z =$	1	2	3	4	5	6
$y=1$	2	3	4	5	6	7
$y=2$	3	4	5	6	7	8
$y=3$	4	5	6	7	8	9
$y=4$	5	6	7	8	9	10
$y=5$	6	7	8	9	10	11
$y=6$	7	8	9	10	11	12

Definition (conditional pmf)

Let $S_Y = \{y : p_Y(y) > 0\}$. The **conditional pmf of X given Y** is

$$p_{X|Y} : \mathbb{R} \times S_Y \rightarrow [0, 1], \quad p_{X|Y}(x \mid y) = \mathbb{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)};$$

the first argument x is a value of X , the second argument y names the row.

Restrict to row y and rescale its probabilities so that they total 1.

Conditional Expectation Given $Y = y$: The Row Average

Definition (discrete case)

Let $S_Y = \{y : p_Y(y) > 0\}$. The **conditional expectation of X given $Y = y$** is the function

$$g: S_Y \rightarrow \mathbb{R}, \quad g(y) = \mathbb{E}[X \mid Y = y] = \sum_x x p_{X|Y}(x \mid y).$$

The ordinary expected value, computed with the row's conditional distribution in place of the whole table's.

What each symbol is, and what the object is not

- x : runs over the values of X , summed away. y : held fixed, names the row. Result: depends on y alone.
- $g: S_Y \rightarrow \mathbb{R}$: a function on the values of Y , not on Ω . Not a random variable.
- $y \notin S_Y$: undefined. A row of probability 0 has no conditional distribution.
- $\mathbb{E}[X \mathbf{1}_{\{Y=y\}}] = \sum_x x p_{X,Y}(x, y) = \mathbb{E}[X \mid Y = y] \mathbb{P}(Y = y)$, with $\mathbf{1}_{\{Y=y\}}: \Omega \rightarrow \{0, 1\}$ the indicator of the row: the row's total, not its average.

Two Dice: The Row Average as a Function of y

$\mathbb{E}[X \mid Y = y]$ for the sum of two dice

Row y contains $y + 1, \dots, y + 6$, each with conditional probability $\frac{1}{6}$:

$$\mathbb{E}[X \mid Y = y] = \frac{1}{6}((y + 1) + \dots + (y + 6)) = y + 3.5.$$

As a function on $S_Y = \{1, \dots, 6\}$:

y	1	2	3	4	5	6
$g(y) = \mathbb{E}[X \mid Y = y]$	4.5	5.5	6.5	7.5	8.5	9.5

each row, then its average

$y=1$	2	3	4	5	6	7	$\rightarrow 4.5$
$y=2$	3	4	5	6	7	8	$\rightarrow 5.5$
$y=3$	4	5	6	7	8	9	$\rightarrow 6.5$
$y=4$	5	6	7	8	9	10	$\rightarrow 7.5$
$y=5$	6	7	8	9	10	11	$\rightarrow 8.5$
$y=6$	7	8	9	10	11	12	$\rightarrow 9.5$

- $g: \{1, \dots, 6\} \rightarrow \mathbb{R}$, $g(y) = y + 3.5$. Six numbers, one per row.
- $g(y) - y = 3.5 = \mathbb{E}[Z]$ for every y : on row y the first die is fixed at y and all the remaining randomness is the second die. Slide 9 makes this a method.

The Continuous Case: Rows Become Slices

Definition (continuous case)

Let X, Y have joint pdf $f_{X,Y}: \mathbb{R}^2 \rightarrow [0, \infty)$, let $S_Y = \{y : f_Y(y) > 0\}$, and define the **conditional pdf of X given Y**

$$f_{X|Y}: \mathbb{R} \times S_Y \rightarrow [0, \infty), \quad f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)},$$

x the value of X , y the slice. The **conditional expectation of X given $Y = y$** is the function

$$g: S_Y \rightarrow \mathbb{R}, \quad g(y) = \mathbb{E}[X | Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx.$$

The row $\{Y = y\}$ is now a horizontal line through the plane, and the slice of $f_{X,Y}$ along it, rescaled to area 1, plays the part of the row's conditional pmf.

- For fixed $y \in S_Y$, $f_{X|Y}(\cdot | y)$ is a pdf in x : $\int f_{X|Y}(x | y) dx = f_Y(y)/f_Y(y) = 1$.
- $\mathbb{P}(Y = y) = 0$, so $\mathbb{P}(\cdot | Y = y)$ cannot be defined by dividing probabilities; the ratio of densities is the limit of conditioning on $\{y \leq Y \leq y + \delta\}$ as $\delta \rightarrow 0$.
- Otherwise nothing changes: fix y , take the slice, rescale, average.

The Two-Stage Point: Horizontal and Vertical Slices

Y uniform on $[0, 1]$; given $Y = y$, X uniform on $[0, y]$. So $f_Y = 1$ on $(0, 1]$, $f_{X|Y}(x | y) = \frac{1}{y}$ on $[0, y]$, and $f_{X,Y}(x, y) = f_{X|Y}(x | y)f_Y(y) = \frac{1}{y}$ on the triangle $0 < x < y < 1$.

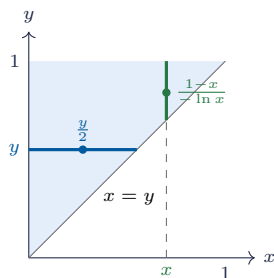
Given $Y = y$: a horizontal slice

$S_Y = (0, 1]$, the slice is uniform on $[0, y]$, and $\mathbb{E}[X | Y = y] = \int_0^y x \frac{1}{y} dx = \frac{y}{2}$.

Given $X = x$: a vertical slice

$f_X(x) = \int_x^1 \frac{1}{y} dy = -\ln x$ on $S_X = (0, 1)$, so $f_{Y|X}(y | x) = \frac{1/y}{-\ln x}$ for $x < y < 1$ and $\mathbb{E}[Y | X = x] = \int_x^1 \frac{y}{y} \frac{dy}{-\ln x} = \frac{1-x}{-\ln x}$.

- The vertical slice is not uniform, $f_{Y|X}(y | x) \propto \frac{1}{y}$ favours small y , so $\mathbb{E}[Y | X = x]$ lies below the slice's midpoint $\frac{1+x}{2}$: at $x = \frac{1}{2}$, 0.7213 against 0.75.



Blue: $\{Y = y\}$, uniform on $[0, y]$.
Green: $\{X = x\}$, density $\propto \frac{1}{y}$ on $[x, 1]$.

Functions of X , and What Is Constant on a Row

The same recipe applies to any random variable, and in particular to any function of X : for $W: \Omega \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E}[W \mid Y = y] = \sum_w w p_{W|Y}(w \mid y), \quad \mathbb{E}[h(X) \mid Y = y] = \sum_x h(x) p_{X|Y}(x \mid y),$$

with $p_{W|Y}$ defined as on slide 4 with W in place of X ; the second formula is the rule $\mathbb{E}[h(X)] = \sum_x h(x)p_X(x)$ applied to the row's pmf.

Observation: on row y , Y is the constant y

Anything that depends on the outcome only through Y is constant on the row. Given $Y = y$, $X = Y + Z$ is $y + Z$: the row's randomness is all in Z .

Two dice: $\mathbb{E}[X^2 \mid Y = y]$ without listing the row

On row y , $X^2 = (y + Z)^2 = y^2 + 2yZ + Z^2$ with y a constant and Z uniform on $\{1, \dots, 6\}$, so

$$\mathbb{E}[X^2 \mid Y = y] = y^2 + 2y\mathbb{E}[Z] + \mathbb{E}[Z^2] = y^2 + 7y + \frac{91}{6}.$$

Head-on you would square six numbers and average, for each of six rows.

- The observation is the seed of the property “taking out what is known” (slide 20).

Total Expectation: The Average of the Row Averages

The rows partition the table. The overall average of the 36 cells can be computed row by row:

$$\mathbb{E}[X] = \sum_{y=1}^6 \underbrace{\frac{1}{6}}_{\mathbb{P}(Y=y)} \cdot \underbrace{(y + 3.5)}_{\text{row average}} = 7.$$

2	3	4	5	6	7	→ 4.5
3	4	5	6	7	8	→ 5.5
4	5	6	7	8	9	→ 6.5
5	6	7	8	9	10	→ 7.5
6	7	8	9	10	11	→ 8.5
7	8	9	10	11	12	→ 9.5

$$\frac{1}{6} \sum \downarrow = 7$$

Theorem (law of total expectation)

discrete:
$$\mathbb{E}[X] = \sum_{y \in S_Y} \mathbb{E}[X \mid Y = y] p_Y(y),$$

continuous:
$$\mathbb{E}[X] = \int_{S_Y} \mathbb{E}[X \mid Y = y] f_Y(y) dy.$$

Average each row, then average the row averages weighted by the row probabilities; because the rows cover the table exactly once, this is the overall average.

Total Expectation: Proof

Proof (discrete case)

$$\begin{aligned}\sum_y \mathbb{E}[X \mid Y = y] p_Y(y) &= \sum_y \left(\sum_x x p_{X|Y}(x \mid y) \right) p_Y(y) && \text{definition} \\ &= \sum_y \sum_x x p_{X,Y}(x, y) && \text{undo the rescaling} \\ &= \sum_x x \sum_y p_{X,Y}(x, y) = \sum_x x p_X(x) = \mathbb{E}[X]. && \text{swap sums; marginal} \quad \square\end{aligned}$$

Line two undoes the rescaling that defined the conditional pmf, and line three sums the cells column-first instead of row-first.

- The swap of the two sums is free on the finite table; in general it needs $\mathbb{E}|X| < \infty$.
- Continuous case: identical, with f for p and $\int$ for $\sum$:

$$\int_{S_Y} \left(\int x f_{X|Y}(x \mid y) dx \right) f_Y(y) dy = \iint x f_{X,Y}(x, y) dx dy = \int x f_X(x) dx.$$

- Any $W: \Omega \rightarrow \mathbb{R}$ in place of X : $\mathbb{E}[W] = \sum_y \mathbb{E}[W \mid Y = y] p_Y(y)$, in particular for $W = h(X)$.

Total Expectation at Work (1): Problem A, a Random Number of Coins

Roll a die, getting $N: \Omega \rightarrow \{1, \dots, 6\}$. Toss N fair coins, independent of the die; $H: \Omega \rightarrow \{0, \dots, 6\}$ is the number of heads. What is $\mathbb{E}[H]$?

By conditioning

Given $N = n$, H is binomial with n tosses (the coins are independent of the die, so the row $\{N = n\}$ does not change their distribution), so $\mathbb{E}[H \mid N = n] = \frac{n}{2}$. Then

$$\mathbb{E}[H] = \sum_{n=1}^6 \frac{n}{2} \cdot \frac{1}{6} = \frac{1}{2} \mathbb{E}[N] = 1.75.$$

Head-on, for comparison

$p_H(h) = \sum_{n=\max(h,1)}^6 \frac{1}{6} \binom{n}{h} 2^{-n}$ for $h = 0, \dots, 6$: seven sums (e.g. $p_H(0) = \frac{63}{384}$, $p_H(1) = \frac{120}{384}$), then $\sum_h h p_H(h)$.

- Given the row $N = n$ the problem is a textbook one. The distribution of H , a mixture of six binomials, was never needed.

Total Expectation at Work (2): The Two-Stage Point, and Variance

Y uniform on $[0, 1]$; given $Y = y$, X uniform on $[0, y]$. What is $\mathbb{E}[X]$?

By conditioning

$$\mathbb{E}[X \mid Y = y] = \frac{y}{2} \text{ and } f_Y = 1 \text{ on } (0, 1], \text{ so } \mathbb{E}[X] = \int_0^1 \frac{y}{2} \cdot 1 \, dy = \frac{1}{4}.$$

Head-on, for comparison

First find the pdf of X by integrating out y : $f_X(x) = \int_x^1 f_{X,Y}(x, y) \, dy = \int_x^1 \frac{dy}{y} = -\ln x$ for $0 < x < 1$. Then $\mathbb{E}[X] = \int_0^1 x(-\ln x) \, dx$, which needs integration by parts, and gives $\frac{1}{4}$.

Definition (variance)

For $X: \Omega \rightarrow \mathbb{R}$ with $\mathbb{E}[X^2] < \infty$: $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

The average squared distance of X from its mean; computing it needs $\mathbb{E}[X^2]$, which total expectation supplies row by row.

- Here $\mathbb{E}[X^2 \mid Y = y] = \frac{y^2}{3}$, so $\mathbb{E}[X^2] = \int_0^1 \frac{y^2}{3} \, dy = \frac{1}{9}$ and $\text{Var}(X) = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}$. Neither computation needed f_X .

$\mathbb{E}[X | Y]$: Paint Every Row With Its Average

Slide 6 gave a function on the values of Y :

$g: S_Y \rightarrow \mathbb{R}$, $g(y) = \mathbb{E}[X | Y = y]$.

Now write $g(y)$ into *every cell of row y* . The result is a new number in each cell of the table, that is, a new random variable.

Definition

The **conditional expectation of X given Y** is the random variable

$$\mathbb{E}[X | Y]: \Omega \rightarrow \mathbb{R},$$

$$\mathbb{E}[X | Y](\omega) = g(Y(\omega)) = \mathbb{E}[X | Y = Y(\omega)].$$

At outcome ω , look up which row ω is in and report that row's average.

- $g(Y)$ is defined on the event $\{Y \in S_Y\}$, which has probability 1; set it to 0 elsewhere. On the two-dice Ω every outcome has positive probability, so there is nothing to fix.

$$\mathbb{E}[X | Y](\omega) = Y(\omega) + 3.5$$

$z =$	1	2	3	4	5	6
$y=1$	4.5	4.5	4.5	4.5	4.5	4.5
$y=2$	5.5	5.5	5.5	5.5	5.5	5.5
$y=3$	6.5	6.5	6.5	6.5	6.5	6.5
$y=4$	7.5	7.5	7.5	7.5	7.5	7.5
$y=5$	8.5	8.5	8.5	8.5	8.5	8.5
$y=6$	9.5	9.5	9.5	9.5	9.5	9.5

Three Tables, Three Levels of Knowledge

	X					
$z=$	1	2	3	4	5	6
$y=1$	2	3	4	5	6	7
$y=2$	3	4	5	6	7	8
$y=3$	4	5	6	7	8	9
$y=4$	5	6	7	8	9	10
$y=5$	6	7	8	9	10	11
$y=6$	7	8	9	10	11	12

you see the outcome

	$\mathbb{E}[X Y]$					
$z=$	1	2	3	4	5	6
$y=1$	4.5	4.5	4.5	4.5	4.5	4.5
$y=2$	5.5	5.5	5.5	5.5	5.5	5.5
$y=3$	6.5	6.5	6.5	6.5	6.5	6.5
$y=4$	7.5	7.5	7.5	7.5	7.5	7.5
$y=5$	8.5	8.5	8.5	8.5	8.5	8.5
$y=6$	9.5	9.5	9.5	9.5	9.5	9.5

you see the row only

	$\mathbb{E}[X]$					
$z=$	1	2	3	4	5	6
$y=1$	7	7	7	7	7	7
$y=2$	7	7	7	7	7	7
$y=3$	7	7	7	7	7	7
$y=4$	7	7	7	7	7	7
$y=5$	7	7	7	7	7	7
$y=6$	7	7	7	7	7	7

you see nothing

Each table is the previous one averaged over what you do not know.

- $\mathbb{E}[X | Y = y]$: a number for each y . $\mathbb{E}[X | Y] = g(Y)$: a random variable, constant on each row $\{Y = y\}$, where its value is $\mathbb{E}[X | Y = y]$.

$\mathbb{E}[X | Y]$: Extreme Cases and the Continuous Case

- $\mathbb{E}[X | X] = X$. Condition on X itself: each “row” $\{X = x\}$ contains only cells where $X = x$, so its average is x . Full knowledge changes nothing.
- $\mathbb{E}[X | Y] = \mathbb{E}[X]$ if Y is **constant**. One row, the whole table. No knowledge.
- $\mathbb{E}[Y | Y] = Y$, and more generally $\mathbb{E}[h(Y) | Y] = h(Y)$ for $h: \mathbb{R} \rightarrow \mathbb{R}$: on row y , $h(Y)$ is the constant $h(y)$, whose average is $h(y)$.

Continuous case

The same definition, $\mathbb{E}[X | Y] = g(Y)$ with $g: S_Y \rightarrow \mathbb{R}$, $g(y) = \int x f_{X|Y}(x | y) dx$. The “rows” $\{Y = y\}$ each have probability 0, but $g(Y): \Omega \rightarrow \mathbb{R}$ is an ordinary random variable: a function of Y .

Examples

Two-stage point: $g(y) = y/2$, so $\mathbb{E}[X | Y] = Y/2$. At an outcome where $Y = 0.8$, the average for X is 0.4, whatever X actually is. Likewise $\mathbb{E}[Y | X] = (1 - X)/(-\ln X)$.

Random coins: $\mathbb{E}[H | N] = N/2$, a random variable with values $\frac{1}{2}, 1, \dots, 3$, each with probability $\frac{1}{6}$.

Property 1: Tower. The Painted Table Has the Same Average

Painting a row with its average does not change the row's total. So the painted table has the same overall average as the original:

Theorem (tower property)

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].$$

Proof

$\mathbb{E}[X \mid Y] = g(Y)$, and the expectation of a function of Y is $\mathbb{E}[g(Y)] = \sum_y g(y) p_Y(y) = \sum_y \mathbb{E}[X \mid Y = y] p_Y(y) = \mathbb{E}[X]$ by total expectation. $\square$

It is the law of total expectation with the y 's hidden inside a random variable.

- That makes it composable: $\mathbb{E}[X \mid Y]$ is itself a random variable, and can be attacked by conditioning on something else.

2	3	4	5	6	7
3	4	5	6	7	8
4	5	6	7	8	9
5	6	7	8	9	10
6	7	8	9	10	11
7	8	9	10	11	12

average 7



4.5	4.5	4.5	4.5	4.5	4.5
5.5	5.5	5.5	5.5	5.5	5.5
6.5	6.5	6.5	6.5	6.5	6.5
7.5	7.5	7.5	7.5	7.5	7.5
8.5	8.5	8.5	8.5	8.5	8.5
9.5	9.5	9.5	9.5	9.5	9.5

average 7

Property 1: Why the Tower Matters

To compute $\mathbb{E}[X]$, compute $\mathbb{E}[X \mid Y]$ for a well-chosen Y and average that instead.

problem	$\mathbb{E}[X \mid Y]$, a formula in Y	then $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$
coins, N from a die	$\mathbb{E}[H \mid N] = N/2$	$\mathbb{E}[N]/2 = 1.75$
two-stage point	$\mathbb{E}[X \mid Y] = Y/2$	$\mathbb{E}[Y]/2 = 1/4$
random sum $S = X_1 + \cdots + X_N$	$\mathbb{E}[S \mid N] = \mu N$	$\mu \mathbb{E}[N]$
waiting time T for HH	$\mathbb{E}[T \mid Y]$ for the opening Y	an equation for $\mathbb{E}[T]$

- In each row X has an unpleasant distribution; $\mathbb{E}[X \mid Y]$ is a one-line formula in Y .
- Choosing Y is the whole art: the row problem must be textbook, and p_Y or f_Y must be known.
- The last two rows are the applications at the end of the lecture.

Property 2: A Problem. Are N and H Correlated?

Problem A again: N the die, H the number of heads among N coins. More coins should mean more heads. How strongly are N and H linked?

Definition (covariance)

For $X, Y: \Omega \rightarrow \mathbb{R}$ with finite second moments:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]; \quad \text{Cov}(X, X) = \text{Var}(X).$$

Positive when X above its mean tends to come with Y above its mean; zero for independent X, Y .

Head-on

$\mathbb{E}[NH] = \sum_{n=1}^6 \sum_{h=0}^n n h p_{N,H}(n, h)$ with $p_{N,H}(n, h) = \frac{1}{6} \binom{n}{h} 2^{-n}$: 21 nonzero terms of the joint pmf, each a binomial coefficient over a power of two.

What conditioning offers

Tower: $\mathbb{E}[NH] = \mathbb{E}[\mathbb{E}[NH | N]]$. On the row $\{N = n\}$ the factor N is the number n . If a constant factor can be pulled out of a row average, then $\mathbb{E}[NH | N = n] = n \mathbb{E}[H | N = n] = n \cdot \frac{n}{2}$, and $\mathbb{E}[NH]$ is an ordinary expectation of a function of N . Slide 9 made this move one row at a time; the next slide states it once, for all rows.

Property 2: Taking Out What Is Known

On row y , anything that is a function of Y is the constant $h(y)$, and a constant factor comes out of an average.

Theorem

For any $h: \mathbb{R} \rightarrow \mathbb{R}$ with $\mathbb{E}|h(Y)X| < \infty$: $\mathbb{E}[h(Y)X \mid Y] = h(Y) \mathbb{E}[X \mid Y]$. In particular $\mathbb{E}[h(Y) \mid Y] = h(Y)$.

Proof (discrete case)

Fix $y \in S_Y$.

$\mathbb{E}[h(Y)X \mid Y = y] = \sum_x h(y) x p_{X|Y}(x \mid y) = h(y) \sum_x x p_{X|Y}(x \mid y) = h(y) \mathbb{E}[X \mid Y = y]$. This holds for every y ; substitute $y = Y(\omega)$. Continuous case identical, with f for p and $\int$ for $\sum$. $\square$

Whatever is already determined by the row is a constant on the row and can be taken out of the row average.

Linearity

$\mathbb{E}[aX + bX' \mid Y] = a \mathbb{E}[X \mid Y] + b \mathbb{E}[X' \mid Y]$ for constants a, b , for the same reason ordinary expectation is linear: row by row. With the theorem: $\mathbb{E}[h(Y) + X \mid Y] = h(Y) + \mathbb{E}[X \mid Y]$.

Property 2: Back to $\text{Cov}(N, H)$, and $\mathbb{E}[XY]$ for Two Dice

$\text{Cov}(N, H)$ for Problem A

$$\mathbb{E}[NH] = \mathbb{E}[\mathbb{E}[NH \mid N]] = \mathbb{E}[N \mathbb{E}[H \mid N]] = \mathbb{E}\left[N \cdot \frac{N}{2}\right] = \frac{1}{2} \mathbb{E}[N^2] = \frac{1}{2} \cdot \frac{91}{6} = \frac{91}{12},$$

so $\text{Cov}(N, H) = \frac{91}{12} - \frac{7}{2} \cdot \frac{7}{4} = \frac{35}{24} \approx 1.46$. With $\text{Var}(N) = \mathbb{E}[N^2] - \mathbb{E}[N]^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$.
 $\text{Cov}(N, H) / \text{Var}(N) = \frac{1}{2}$, the slope of $\mathbb{E}[H \mid N] = \frac{1}{2}N$.

$\mathbb{E}[XY]$ for two dice, without listing 36 products

$$\mathbb{E}[XY] = \mathbb{E}[\mathbb{E}[XY \mid Y]] = \mathbb{E}[Y \mathbb{E}[X \mid Y]] = \mathbb{E}[Y(Y + 3.5)] = \mathbb{E}[Y^2] + 3.5 \mathbb{E}[Y] = \frac{91}{6} + \frac{49}{4} = \frac{329}{12},$$

about 27.42, so $\text{Cov}(X, Y) = \frac{329}{12} - 7 \cdot 3.5 = \frac{35}{12}$.

- Both computations are the same three steps: tower; take out the factor that is a function of the conditioning variable; an ordinary expectation of a function of that variable.
- The 21-term double sum of the previous slide was never written down.

Property 3: Independence

Definition (independence)

$X, Y: \Omega \rightarrow \mathbb{R}$ are **independent**, written $X \perp\!\!\!\perp Y$, if

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B) \quad \text{for all } A, B \subseteq \mathbb{R}.$$

$\xi_1, \dots, \xi_n: \Omega \rightarrow \mathbb{R}$ are **independent** if

$\mathbb{P}(\xi_1 \in A_1, \dots, \xi_n \in A_n) = \prod_{i=1}^n \mathbb{P}(\xi_i \in A_i)$ for all $A_i \subseteq \mathbb{R}$; a sequence, if every finite part is.

The row you are in tells you nothing about which column you are in.

- Two dice: $p_{Y,Z}(y, z) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = p_Y(y) p_Z(z)$; summing over the cells of $\{Y \in A\} \cap \{Z \in B\}$ gives the definition for all A, B . So $Y \perp\!\!\!\perp Z$.

Fact (stated without proof)

Functions of disjoint groups of independent variables are independent: $\xi_1, \xi_2, \dots$ independent implies $f(\xi_1, \dots, \xi_5) \perp\!\!\!\perp g(\xi_6, \xi_7, \xi_8)$ for all $f: \mathbb{R}^5 \rightarrow \mathbb{R}$, $g: \mathbb{R}^3 \rightarrow \mathbb{R}$.

$Z(\omega) = z$: every row alike

$z =$	1	2	3	4	5	6
$y=1$	1	2	3	4	5	6
$y=2$	1	2	3	4	5	6
$y=3$	1	2	3	4	5	6
$y=4$	1	2	3	4	5	6
$y=5$	1	2	3	4	5	6
$y=6$	1	2	3	4	5	6

Property 3: A Problem. A Random Walk, Seen From Halfway

Let $\xi_1, \xi_2, \dots : \Omega \rightarrow \{-1, 1\}$ be independent with $\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = \frac{1}{2}$, and $S_n = \xi_1 + \dots + \xi_n$ the position after n steps. You watch the first five steps and see S_5 . What do you expect S_8 to be, and S_8^2 ? That is: find $\mathbb{E}[S_8 \mid S_5]$ and $\mathbb{E}[S_8^2 \mid S_5]$.

Head-on

Row $\{S_5 = s\}$ for $s \in \{-5, -3, -1, 1, 3, 5\}$. On it, S_8 has the conditional pmf $p_{S_8|S_5}(t \mid s) = p_{S_5, S_8}(s, t) / p_{S_5}(s)$, which needs the joint pmf of (S_5, S_8) : count paths. For $s = 3$: $t = 0, 2, 4, 6$ with probabilities $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$, mean 3, second moment 12. Then five more rows.

What conditioning offers

$S_8 = S_5 + D$ with $D = \xi_6 + \xi_7 + \xi_8$. On row $\{S_5 = s\}$ the first term is the constant s and comes out. The second is built from steps the row says nothing about, so the row should not change its distribution: $\mathbb{E}[D \mid S_5]$ should be $\mathbb{E}[D] = 0$.

- “The row says nothing about D ” is $D \perp\!\!\!\perp S_5$, which holds by the fact on the previous slide. The next slide says exactly what it buys.

Property 3: Independence. Every Row Looks Like the Whole Table

Theorem

If $X \perp\!\!\!\perp Y$ then $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$, a constant; equivalently $\mathbb{E}[X \mid Y = y] = \mathbb{E}[X]$ for every $y \in S_Y$.

Proof (discrete case)

Take $A = \{x\}$, $B = \{y\}$ in the definition: $p_{X,Y}(x, y) = p_X(x) p_Y(y)$. So for every $y \in S_Y$,

$$p_{X|Y}(x \mid y) = \frac{p_X(x) p_Y(y)}{p_Y(y)} = p_X(x) :$$

each row carries the same distribution of X as the whole table. Hence every row average is $\sum_x x p_X(x) = \mathbb{E}[X]$. Continuous case identical, with $f_{X,Y} = f_X f_Y$. □

Knowing the row tells you nothing about X , so the row average is the overall average.

- Two dice: $Z \perp\!\!\!\perp Y$; every row of the Z -table is $1, \dots, 6$, so $\mathbb{E}[Z \mid Y] = 3.5$ on every row.
- Also $\mathbb{E}[h(X) \mid Y] = \mathbb{E}[h(X)]$ for $h: \mathbb{R} \rightarrow \mathbb{R}$, since $h(X) \perp\!\!\!\perp Y$ too ($\{h(X) \in A\} = \{X \in h^{-1}(A)\}$).

Property 3: The Random Walk Done, and a Recipe

$$\mathbb{E}[S_8 \mid S_5] \text{ and } \mathbb{E}[S_8^2 \mid S_5]$$

$D = \xi_6 + \xi_7 + \xi_8$ takes the values $-3, -1, 1, 3$ with probabilities $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$, so $\mathbb{E}[D] = 0$ and $\mathbb{E}[D^2] = 9 \cdot \frac{1}{4} + 1 \cdot \frac{3}{4} = 3$; and $D \perp\!\!\!\perp S_5$ (disjoint groups of steps).

$$\mathbb{E}[S_8 \mid S_5] = \mathbb{E}[S_5 \mid S_5] + \mathbb{E}[D \mid S_5] = S_5 + \mathbb{E}[D] = S_5,$$

$$\mathbb{E}[S_8^2 \mid S_5] = \mathbb{E}[S_5^2 + 2S_5D + D^2 \mid S_5] = S_5^2 + 2S_5 \mathbb{E}[D] + \mathbb{E}[D^2] = S_5^2 + 3.$$

Check on row $s = 3$: $\frac{1}{8} \cdot 0 + \frac{3}{8} \cdot 4 + \frac{3}{8} \cdot 16 + \frac{1}{8} \cdot 36 = 12 = 3^2 + 3$.

The recipe

To compute $\mathbb{E}[X \mid Y]$ without listing rows, write X as sums and products of *functions of Y* and *quantities independent of Y* . The former come out unchanged (taking out what is known); the latter are replaced by their ordinary expectations (independence).

- Two dice in one line: $X = Y + Z$ with $Z \perp\!\!\!\perp Y$, so $\mathbb{E}[X \mid Y] = Y + \mathbb{E}[Z] = Y + 3.5$; and $\mathbb{E}[X^2 \mid Y] = Y^2 + 2Y \mathbb{E}[Z] + \mathbb{E}[Z^2]$.
- $\mathbb{E}[S_8 \mid S_5] = S_5$: from wherever the walk stands, its expected future position is where it is. This is the martingale property, taken up in the stochastic-processes lectures.

Property 3: What Independence Does Not Say

The converse fails: $\mathbb{E}[X | Y] = \mathbb{E}[X]$ does not force $X \perp\!\!\!\perp Y$

Y uniform on $\{-1, 0, 1\}$ and $W = Y^2$. Conditioning on W creates two rows:

row	outcomes	values of Y	row average
$\{W = 0\}$	$\{0\}$	0	0
$\{W = 1\}$	$\{-1, 1\}$	-1, 1	0

So $\mathbb{E}[Y | W] = 0 = \mathbb{E}[Y]$, yet W is a function of Y :

$$\mathbb{P}(Y = 0, W = 0) = \frac{1}{3} \neq \mathbb{P}(Y = 0)\mathbb{P}(W = 0) = \frac{1}{9}.$$

- Equal row *averages* do not mean equal row *distributions*. Independence is the second; the theorem only needs, and $\mathbb{E}[X | Y] = \mathbb{E}[X]$ only delivers, the first.
- What $\mathbb{E}[X | Y] = \mathbb{E}[X]$ does deliver, by tower and taking out:
 $\mathbb{E}[h(Y)X] = \mathbb{E}[h(Y)\mathbb{E}[X | Y]] = \mathbb{E}[h(Y)]\mathbb{E}[X]$, i.e. $\text{Cov}(h(Y), X) = 0$ for every $h: \mathbb{R} \rightarrow \mathbb{R}$.
Here $\text{Cov}(W, Y) = \mathbb{E}[Y^3] - \mathbb{E}[W]\mathbb{E}[Y] = 0$.

Properties: Summary

property	statement	one-glance reason
tower	$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$	painting rows with their averages keeps the total
linearity	$\mathbb{E}[aX + bX' \mid Y] = a\mathbb{E}[X \mid Y] + b\mathbb{E}[X' \mid Y]$	averages are linear, row by row
taking out what is known	$\mathbb{E}[h(Y)X \mid Y] = h(Y)\mathbb{E}[X \mid Y]$	$h(Y)$ is constant on a row
known variables	$\mathbb{E}[h(Y) \mid Y] = h(Y), \quad \mathbb{E}[X \mid X] = X$	a constant's average is itself
independence	$X \perp\!\!\!\perp Y \Rightarrow \mathbb{E}[X \mid Y] = \mathbb{E}[X]$	every row has the table's distribution of X

object	type	meaning
$\mathbb{E}[X]$	number	average over the whole table
$\mathbb{E}[X \mid Y = y]$	number for each $y \in S_Y$; a function $g: S_Y \rightarrow \mathbb{R}$	average over row y
$\mathbb{E}[X \mid Y]$	random variable $g(Y): \Omega \rightarrow \mathbb{R}$	row average, written into every cell of the row

Application 1: Random Sums. Problem B

Let $N: \Omega \rightarrow \{0, 1, 2, \dots\}$ and let $X_1, X_2, \dots: \Omega \rightarrow \mathbb{R}$ have $\mathbb{E}[X_i] = \mu$ for all i . Put

$$S = X_1 + X_2 + \dots + X_N \quad (S = 0 \text{ when } N = 0).$$

Two layers of randomness: how many terms, and what each term is.

Problem B, and why it is hard head-on

N claims, $\mathbb{E}[N] = 20$; claim i has size X_i with $\mathbb{E}[X_i] = \mathcal{L}1500$. Find $\mathbb{E}[S]$. The pmf or pdf of S is $\sum_n \mathbb{P}(N = n) \cdot (\text{distribution of } X_1 + \dots + X_n)$, an n -fold convolution for each n ; and here neither the distribution of N nor that of the X_i is even specified. Only the two means are.

What conditioning offers

On the row $\{N = n\}$, $S = X_1 + \dots + X_n$ is a sum of a *fixed* number of terms, and $\mathbb{E}[X_1 + \dots + X_n] = n\mu$ by linearity. So $\mathbb{E}[S \mid N = n]$ would be $n\mu$, provided that restricting to the row does not change the distribution of the terms.

- That proviso is a genuine assumption about how N is decided. It is stated next.

Application 1: Random Sums. The Assumption

Assumption: N is independent of the X_i

For every n and all sets $A_1, \dots, A_n \subseteq \mathbb{R}$,

$$\mathbb{P}(N = n, X_1 \in A_1, \dots, X_n \in A_n) = \mathbb{P}(N = n) \mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n).$$

Given $N = n$, the joint distribution of $(X_1, \dots, X_n)$ is what it was unconditionally.

- Not assumed: independence of the X_i from each other, or identical distributions. Only equal means μ .
- Holds when N is settled by a separate mechanism: the number of customers before anyone spends; the number of claims regardless of their sizes.
- Fails when N is decided by looking at the X_i : “stop at the first six”; “include the second term only if it is small” (slide 31).

Application 1: Random Sums. Theorem and Proof

Theorem

Under the assumption, $\mathbb{E}[S \mid N = n] = n\mu$ for every n , hence $\mathbb{E}[S \mid N] = \mu N$ and $\mathbb{E}[S] = \mu \mathbb{E}[N]$.

Proof

On the row $\{N = n\}$, $S = W_n := X_1 + \cdots + X_n$. By the assumption, the row $\{N = n\}$ does not change the distribution of W_n , so Property 3, applied to W_n at the row n , gives

$$\mathbb{E}[S \mid N = n] = \mathbb{E}[W_n \mid N = n] = \mathbb{E}[W_n] = n\mu$$

by linearity. Substitute $n = N$ to get $\mathbb{E}[S \mid N] = \mu N$; then $\mathbb{E}[S] = \mathbb{E}[\mu N] = \mu \mathbb{E}[N]$ by the tower property. □

Given the row, the random number of terms is a fixed number and linearity does the rest.

- The three steps again: condition on the layer that makes the problem hard; on the row it is gone; tower.
- Nothing about the distribution of S was computed. Its mean is all that was asked for and all that was found.

Application 1: Back to the Insurer, and When the Assumption Fails

Problem B

$\mathbb{E}[S] = \mu \mathbb{E}[N] = 1500 \cdot 20 = \pounds 30\,000$, whatever the distributions of N and of the claim sizes, as long as the count is independent of the sizes.

N decided by peeking at a term: the formula gives the wrong number

Two dice X_1, X_2 . Let $N = 1$ if $X_2 = 6$ and $N = 2$ otherwise (the second die counts only if it is not a six). Then $S = X_1 + X_2 \mathbf{1}_{\{X_2 \neq 6\}}$ and

$$\mathbb{E}[S] = 3.5 + \frac{1+2+3+4+5}{6} = 6, \quad \text{but} \quad \mu \mathbb{E}[N] = 3.5 \cdot \frac{11}{6} = \frac{77}{12} \approx 6.42.$$

On the row $\{N = 2\} = \{X_2 \neq 6\}$ the second term averages 3, not $\mu = 3.5$: the row changed the distribution of a term.

- “Stop at the first six” also violates the assumption, yet $\mu \mathbb{E}[N] = 3.5 \cdot 6 = 21$ is right. That is Wald’s identity, a separate theorem: its hypothesis is that N never looks *ahead*. The peeking example looks ahead.

Application 2: Waiting for HH. Problem C

Toss a fair coin repeatedly. Let $T: \Omega \rightarrow \{2, 3, \dots\}$ be the number of the toss on which two consecutive heads are first completed: for THTHH..., $T = 5$. We want $\mathbb{E}[T]$.

Why head-on is hard

$\mathbb{P}(T = n)$ is the number of H/T strings of length n that end in HH and contain no earlier HH, over 2^n . The count is the Fibonacci number F_{n-1} , so $\mathbb{P}(T = n) = F_{n-1}/2^n$, and $\mathbb{E}[T] = \sum_{n \geq 2} n F_{n-1}/2^n$ needs generating functions to sum.

The rows we condition on here are described by events rather than by the value of a named variable, so:

Definition (conditional expectation given an event)

For $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$ and $\mathbf{1}_A: \Omega \rightarrow \{0, 1\}$ the indicator of A ,

$$\mathbb{E}[X \mid A] = \sum_x x \mathbb{P}(X = x \mid A) = \mathbb{E}[X \mid \mathbf{1}_A = 1].$$

A row can be any event of positive probability: restrict to A , rescale, average.

Application 2: Waiting for HH. The Restart Property

Let A_k be the event that the first k tosses form a given H/T string ending in T; $\mathbb{P}(A_k) = 2^{-k} > 0$. On A_k no progress towards HH has been made at toss k .

Restart property

Let T' be the number of tosses after toss k until HH is first completed among tosses $k+1, k+2, \dots$. Then on A_k , $T = k + T'$, and

$$\mathbb{E}[T \mid A_k] = k + \mathbb{E}[T' \mid A_k] = k + \mathbb{E}[T].$$

Why

T' is a function of tosses $k+1, k+2, \dots$, and $\mathbf{1}_{A_k}$ of tosses $1, \dots, k$: disjoint groups of independent tosses, so $T' \perp\!\!\!\perp \mathbf{1}_{A_k}$ and, by Property 3, $\mathbb{E}[T' \mid A_k] = \mathbb{E}[T']$. The tosses $k+1, k+2, \dots$ are a fair coin tossed repeatedly, exactly the experiment defining T , so T' has the distribution of T and $\mathbb{E}[T'] = \mathbb{E}[T]$. □

On such a row T is k plus a fresh copy of T , so the row average is k plus the very unknown we are after.

Application 2: Waiting for HH. The Computation

The conditioning variable. Let $Y: \Omega \rightarrow \{1, 2, 3\}$ record how the game opens:

$$Y = \begin{cases} 1 & \text{toss 1 is T} \\ 2 & \text{tosses 1,2 are H,T} \\ 3 & \text{tosses 1,2 are H,H} \end{cases}$$

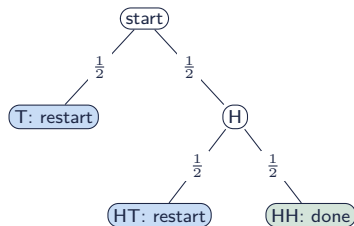
with $p_Y(1) = \frac{1}{2}$, $p_Y(2) = p_Y(3) = \frac{1}{4}$. Every outcome falls in exactly one case: three rows.

Row averages. Rows 1 and 2 are the events $A_1 = \{T\}$ and $A_2 = \{HT\}$ (restart); row 3 is finished:

$$\mathbb{E}[T \mid Y = 1] = 1 + \mathbb{E}[T], \quad \mathbb{E}[T \mid Y = 2] = 2 + \mathbb{E}[T],$$

and $\mathbb{E}[T \mid Y = 3] = 2$. **Total expectation.**

$$\begin{aligned} \mathbb{E}[T] &= \frac{1}{2}(1 + \mathbb{E}[T]) + \frac{1}{4}(2 + \mathbb{E}[T]) + \frac{1}{4} \cdot 2 \\ &= \frac{3}{4}\mathbb{E}[T] + \frac{3}{2}, \quad \text{so } \mathbb{E}[T] = 6. \end{aligned}$$



Two of the three leaves lead back to the start. That is why $\mathbb{E}[T]$ appears on both sides.

Application 2: The Same Move, Elsewhere

Whenever a process restarts after certain outcomes, condition on the first step: the unknown mean appears on both sides of the total-expectation equation, and the equation solves it.

- **Rolls until the first six**, T_6 : $\mathbb{E}[T_6] = \frac{1}{6} \cdot 1 + \frac{5}{6} (1 + \mathbb{E}[T_6])$, so $\mathbb{E}[T_6] = 6$.
- **Waiting for a T**, T_T : the same equation with $\frac{1}{2}$ for $\frac{1}{6}$ gives $\mathbb{E}[T_T] = 2$.
- **Pattern HT**, T_{HT} : after the first H, wait for a T. $\mathbb{E}[T_{HT}] = \frac{1}{2} (1 + \mathbb{E}[T_{HT}]) + \frac{1}{2} (1 + \mathbb{E}[T_T])$, so $\mathbb{E}[T_{HT}] = 4 < 6$: when HT fails (HH), the second H is a head start; when HH fails (HT), nothing is.
- **Pattern HHH**: 14. **Two sixes in a row**: 42. Same equation, more rows.
- **No memory**: $\mathbb{E}[T_6 \mid T_6 > 3] = 3 + \mathbb{E}[T_6] = 9$, by the restart property at $k = 3$.

The equation is legitimate

Solving $\mathbb{E}[T] = \frac{3}{4}\mathbb{E}[T] + \frac{3}{2}$ presumes $\mathbb{E}[T] < \infty$. Each block of two tosses is HH with probability $\frac{1}{4}$, so $\mathbb{P}(T > 2m) \leq (\frac{3}{4})^m$, and $\mathbb{E}[T] = \sum_{n \geq 0} \mathbb{P}(T > n) < \infty$ (tail-sum formula for $\{0, 1, 2, \dots\}$ -valued T ; standard, not proved here).